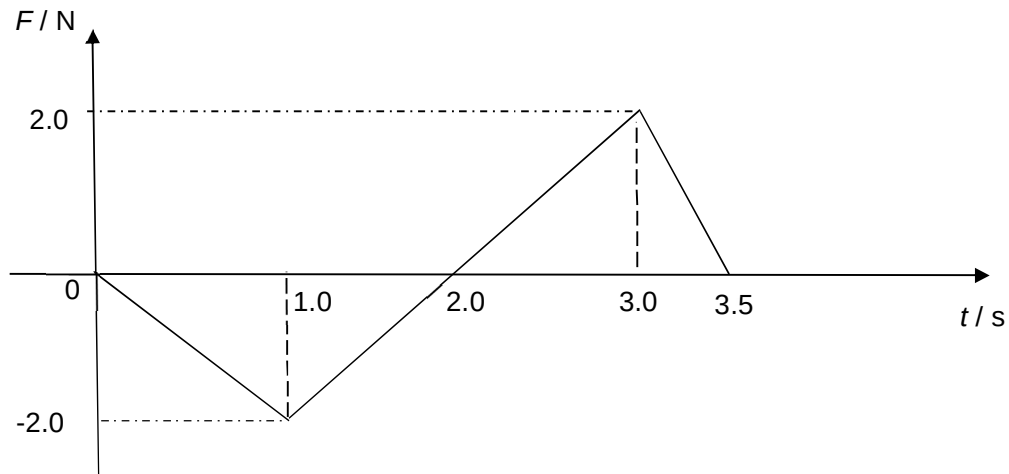


- 5 A resultant force F acts on a mass of 150 g. The variation with time t of F is shown on the diagram below.



At time $t = 0$ s, the mass is moving with a velocity of 2.0 m s^{-1} .

What is the speed of the mass at $t = 3.0$ s?

	A	2.7 m s^{-1}
	B	4.7 m s^{-1}
	C	6.7 m s^{-1}
	D	8.7 m s^{-1}

Answer: B

$$\begin{aligned}
 \text{Change in momentum} &= \text{area under the graph} \\
 &= -\frac{1}{2} (2) (2) + \frac{1}{2} (2) (1) \\
 &= -1.0 \text{ Nm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Therefore, } m \Delta v &= -1.0 \\
 \Delta v &= -1.0 / (150 \times 10^{-3}) \\
 &= -6.67 \text{ m s}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 \text{Hence, } V_f - v_i &= -6.67 \\
 V_f &= -6.67 + 2.0 = -4.67 \text{ m s}^{-1}
 \end{aligned}$$

Therefore the speed of mass = 4.7 m s^{-1} (2 s.f.)